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Elaborate on the following topics with examples:

- Sequential circuits that are not synchronized by a clock
- Analysis of asynchronous sequential circuits
- Hazards that cause incorrect behaviour of a circuit
- Synthesis of asynchronous sequential circuits
- The concept of stable and unstable states
- Hazards that cause incorrect behaviour of a circuit
- Timing issues in digital circuits

• Sequential circuits that are not synchronised by a clock

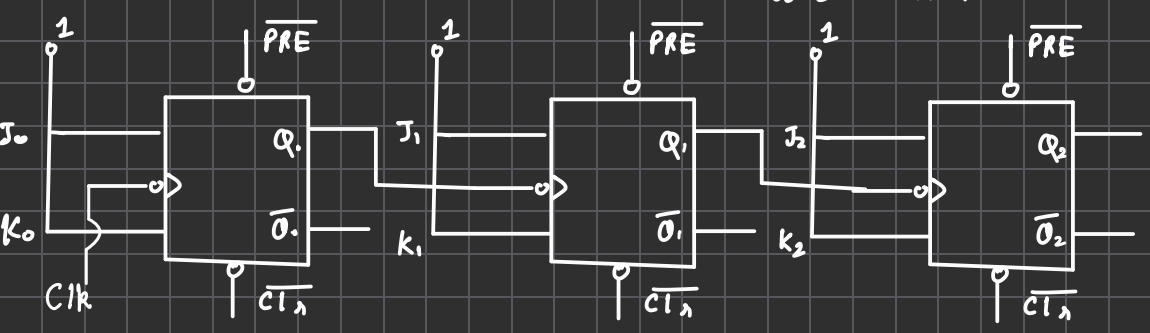
Sequential circuits that are not synchronised by a clock are called Asynchronous sequential circuits.

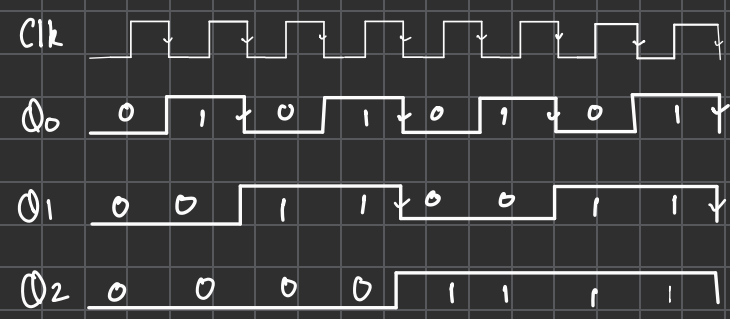
Asynchronous Sequential are circuits that are not synchronised by a common clock signal.

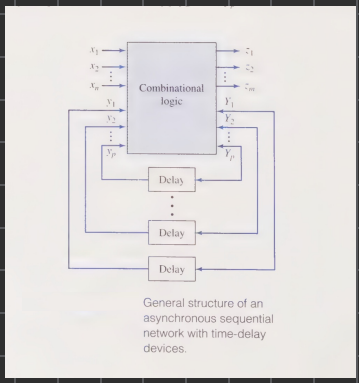
Instead, their outputs change immediately in response to changes in input signals.

Example: Mod 8 counter

Mod 8 counter







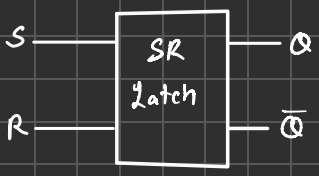
General structure of an asynchronous sequential network with time-delay devices.

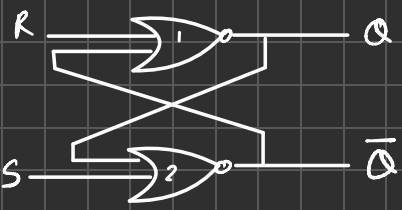
It is asychronous because they dont have a common clock. The clock of next flip flop is the output of previous flip flop. The first flip has an external clock

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• Analysis of Asynchronous Sequential Circuits

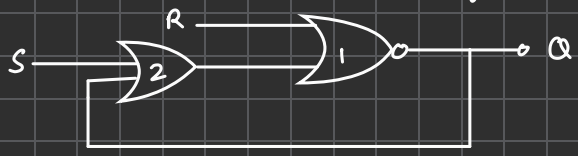
Let take a SR (Set-Reset) Latch for example





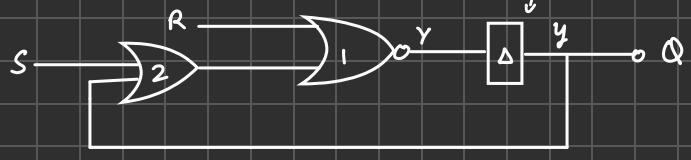
It is an asynchronous circuit because changes in the valve of output Q occur without having to wait for a synchronising clock pulse. In response to a change in either S or R input, the value of Q will change after a short propogation time through the NOR gates.

Redrawing and rearranging circuit



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Introducing propogation delay



By analyzing SR Latch given above, we can derive a state assigned table.

Present state y	Next state SR = 00	01	10	11
	Y	Y	Y	Y
0	(0)	(0)	1	(0)
1	(1)	0	(1)	0

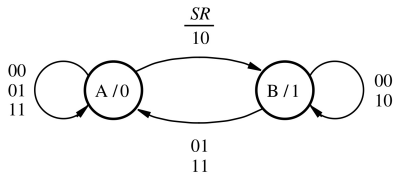
From state assigned table, we can derive state table.

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Present state	Next state SR = 00	01	10	11	Output Q
A	(A)	(A)	B	(A)	0
B	(B)	A	(B)	A	1

(a) State table

Since the output Q depends only on present state, the circuit is Moore-type FSM for which a state diagram can be made.



(b) State diagram

Deriving a minimal POS from state assigned Table

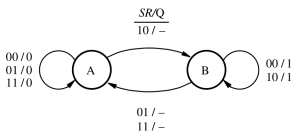
$$\bar{R}Y + S\bar{R} \Rightarrow \bar{R}(S+y) = Y$$

$$\Rightarrow \boxed{Y = \bar{R}\cdot(S+y)}$$

Similarly SR Latch can be represented by Mealy Model.

Present state	Next state SR = 00	01	10	11	Output, Q 00	01	10	11
A	(A)	(A)	B	(A)	0	0	–	0
B	(B)	A	(B)	A	1	–	1	–

(a) State table



(b) State diagram

$$\boxed{Y = y}$$

• Hazards that cause incorrect behaviour of circuit.

Static Hazards:

A static hazard occurs when the output should remain constant, but for a brief unwanted pulse (glitch) appears due to unequal path delays.

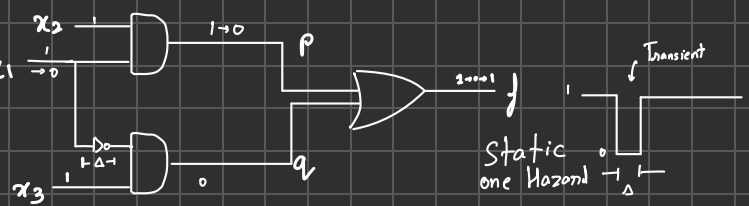
Static - 1 Hazard:

Static-1 Hazard occurs when the output should remain 1, but briefly drops to 0 during an input transition.

Output : $1 \to 0 \to 1$

It occurs when two or more paths produce 1 at the output when an input changes, one path responds faster than the other. So for a brief moment, both paths output 0 and the combined output glitches to 0.

Example: Consider the function $f = x_1x_2 + \bar{x}_1x_3$



Suppose $x_1 = x_2 = x_3 = 1$, then $f = 1$

Let x_1 change from $1 \to 0$. The circuit is supposed to maintain $f = 1$. But the change in x_1 will be observed at point p before q because the path from x_1 to q has an extra gate. (NOT). Thus the signal at P will become 0 before the signal at q becomes equal to 1. For a short time p & q will cause 0, causing f to drop to 0 before it recovers back to 1.

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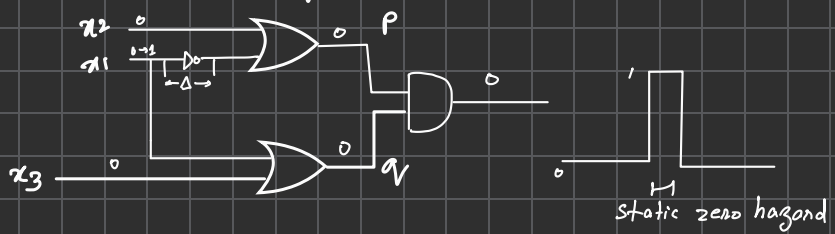
Similarly,

Static - 0 Hazards :

Static-0 Hazard occurs when the output should remain 0, but briefly rises to 1 during an input transition.

Output : 0 → 1 → 0

Consider the function $f = (x_2 + \bar{x_1}).(x_1 + x_3)$

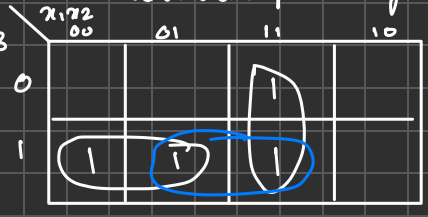


Suppose $x_1 = x_2 = x_3 = 0$, then $f = 0$

Let x_1 change from 0 → 1. The circuit is supposed to maintain $f = 0$ But the change in x_1 will be observed at point q before p because the path from x_1 to p has an extra gate. (NOT). Thus the signal at Q will become 1 before the signal at P becomes equal to 0. For a short time p & q will cause 1, causing f to rise to 1 before it recovers back to 0.

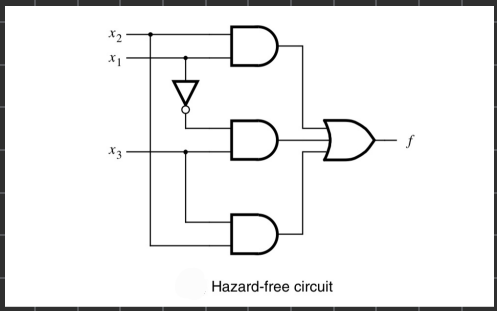
A potential hazard exists whenever two adjacent 1s in a Kmap are not covered by a single product term.

Therefore to eliminate hazard we need to find a cover in which some product term includes pairs of two adjacent 1's.



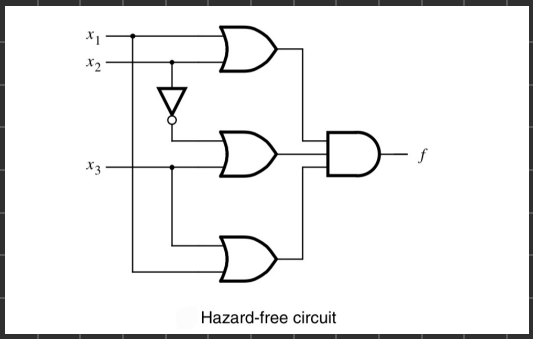
$f = x_1x_2 + \bar{x_1}x_3 + x_2x_3$

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Hazard-free circuit

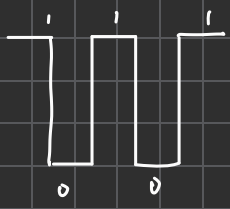
Similarly, we can eliminate hazard in $f = (x_2 + \bar{x}_1).(x_1 + x_3)$



Hazard-free circuit

Dynamic Hazard:

A Dynamic Hazard occurs in a digital circuit when the output is supposed to change only once, (from 0→1 or 1→0) but instead it changes multiple times before setting to its final value.
During a single input change, the output may oscillate like 0→1→0→1 instead of switching cleanly



• Synthesis of Asynchronous sequential circuits.

Problem Definition
↓
State Diagram / Flow Diagram (for an FSM)
↓
Flow table
↓
State Reduction (if required)
↓
State Exitations & State Assignment
↓
Kmap simplification
↓
Circuit

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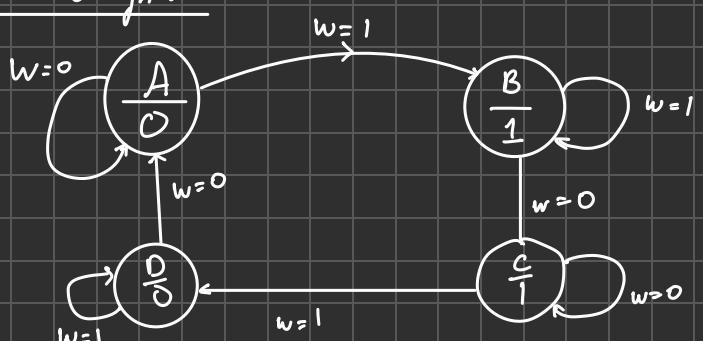
Example.

Problem Statement:

Design a Asynchronous Sequential ckt which gives output=1 when odd no of pulses feed to it serially & gives output = 1
when even no of pulses feed to it serially & gives output = 0



State Diagram



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Flow Table:

PS	NS w=0	NS w=1	O/P z
A	(A)	B	0
B	C	(B)	1
C	(C)	D	1
D	A	(D)	0

Skipping State Reduction

	PS $y_1 y_2$	NS w=0 $Y_1 Y_2$	NS w=1 $Y_1 Y_2$	z
A	00	(00)	01	0
B	01	10	(01)	1
C	10	(10)	11	1
D	11	00	(11)	0

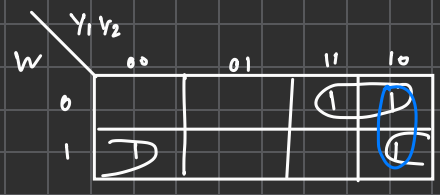
Races

Poor State Assignment

	PS $y_1 y_2$	NS w=0 $Y_1 Y_2$	NS w=1 $Y_1 Y_2$	z
A	00	(00)	01	0
B	01	11	(01)	1
C	11	(11)	10	1
D	10	00	(10)	0

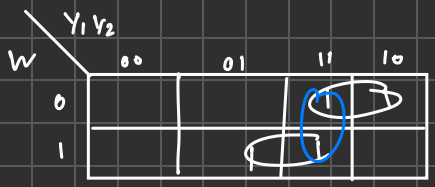
Correct State Evaluation

K map



$$Y_1 = w\bar{y_2} + \bar{w}y_1 + y_1\bar{y_2}$$

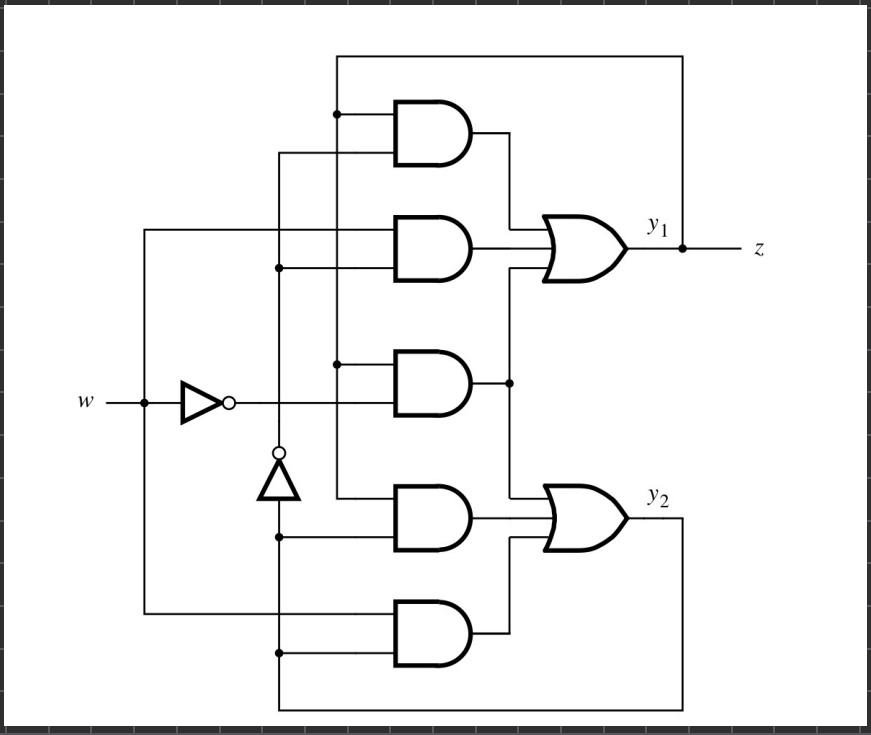
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$$Y_2 = wy_2 + \bar{w}y_1 + y_1y_2$$

$$Z = Y_1$$

Circuit Diagram :



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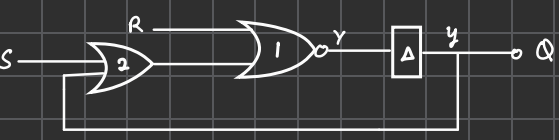
• Concept of Stable and Unstable states.

In asynchronous sequential circuits, the circuits is defined by the present present values of the feedback variables (lets call them y). The combinational part of the circuit generates the next-state variables (lets call them Y).

A circuit is in a stable state when the next state variables are the same as the present state variables (Y=y) for a given set of inputs. The circuit will remain in this state as long as the inputs do not change.
In a flow table, these states are typically circled.

An unstable state exists when the next states variables are different from present state variables (Y≠y). When in an unstable state, the circuit will automatically to the new state (y will become Y) after a propogation delay.

Example: The SR latch



Present state y	Next state SR = 00	01	10	11
	Y	Y	Y	Y
0	(0)	(0)	1	(0)
1	(1)	0	(1)	0

(b) State-assigned table

Stable State: If the present state is y=0 & S=0, R=0, the circuits logic generates a next state of Y=0. Since Y=y, the circuit is stable and will not change.

Unstable State & Transition: If the circuit is in the stable state y=0 with I/P S=0, R=0, and the input S changes to 1, the logic generates a next state of Y=1. Now Y≠y, so state is unstable. After a propogation delay, the y variable will change to 1, bringing the circuit to a new stable state (where y=1, S=1, R=0 & Y=1).

• Hazards that cause incorrect behaviour of circuit.

Static Hazards:

A static hazard occurs when the output should remain constant, but for a brief unwanted pulse (glitch) appears due to unequal path delays.

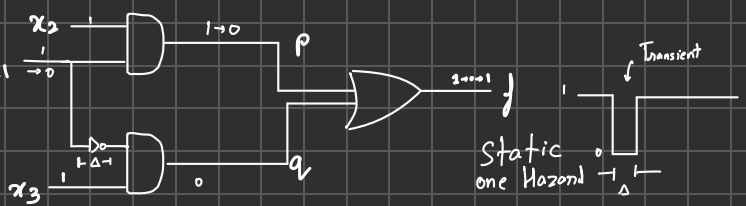
Static - 1 Hazard:

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Output : $1 \to 0 \to 1$

It occurs when two or more paths produce 1 at the output when an input changes, one path responds faster than the other. So for a brief moment, both paths output 0 and the combined output glitches to 0.

Example: Consider the function $f = x_1x_2 + \bar{x}_1x_3$



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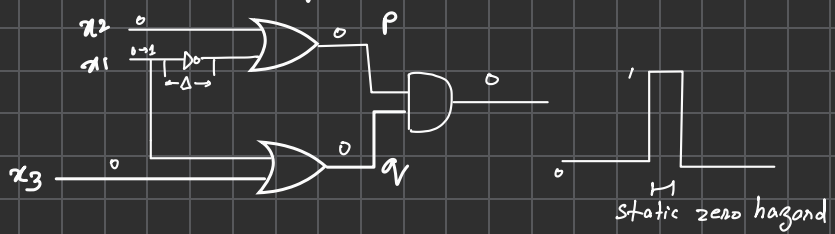
Similarly,

Static - 0 Hazards :

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Consider the function $f = (x_2 + \bar{x_1}).(x_1 + x_3)$

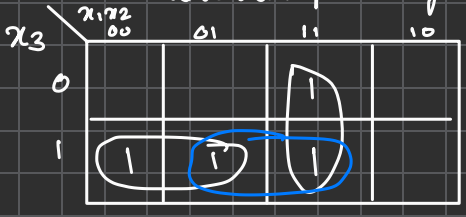


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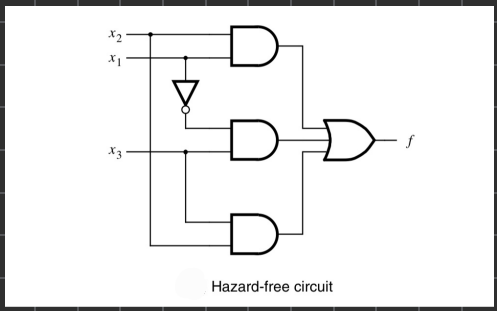
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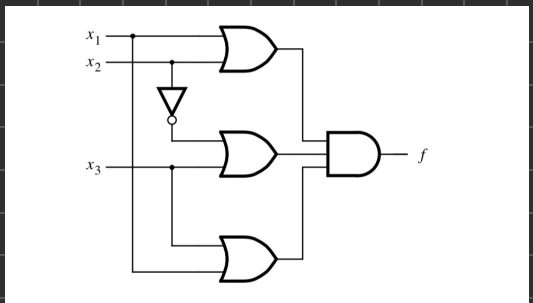
$f = x_1x_2 + \bar{x_1}x_3 + x_2x_3$

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Hazard-free circuit

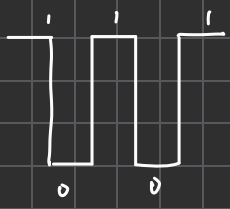
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Hazard-free circuit

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A Dynamic Hazard occurs in a digital circuit when the output is supposed to change only once, (from 0→1 or 1→0) but instead it changes multiple times before setting to its final value.
During a single input change, the output may oscillate like 0→1→0→1 instead of switching cleanly



• Timing Issue in digital circuits.

In asynchronous circuits, timing issue are critical because there is no clock to ensure that all signals are stable before a state change. The most significant timing issues are:

i) Hazards (Discussed above)
ii) Progation delays (tp). : This is the time taken for a logic signal to propogate from an input to an output of a logic gate.

In a synchronous circuit, the maximum propogation delay along the critical path dictates the minimum allowable clock period.

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iii) Race condition

A race condition occurs when a single input change requires two or more state variables to change at the same time. Due to difference in propogation delays, one variable will inevitably change faster than the other, and the order is unpredictable.

Example:

Present state y_2y_1	Next state $w=0$	Next state $w=1$	Output z
	Y_2Y_1		
00	(00)	01	0
01	10	(01)	1
10	(10)	11	1
11	00	(11)	0

(a) Poor state assignment

A transition from state $y_2y_1 = 11$ to 00 is problematic.

If y_1 changes first, the circuit momentarily enters state 10

If y_2 changes first, the circuit momentarily enters state 01.

This race is a critical race, because the two intermediate states lead to different final stable states, making the circuits behaviour unreliable

Race free assignments, such as those where only one state variable changes at a time are used to prevent this.

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